BMS INSTITUTE OF TECHNOLOGY AND MANAGEMENT

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Department of Computer Science and Engineering

Environmental Science Project-BCSP508

Synopsis

Academic Year 2026-27

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Urban Green-Cover Intelligence and Historical Change Analysis System		

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Abstract

Urban green cover moderates local temperature, improves air quality, reduces surface runoff and provides public recreational space, yet it is steadily reduced by urban expansion. Most existing tree and vegetation maps describe only the present condition and therefore cannot show how a city's green cover has evolved. This project proposes the Urban Green-Cover Intelligence and Historical Change Analysis System, a software-based Geographic Information System (GIS) and environmental intelligence platform that analyses multi-year satellite imagery to measure, map and explain changes in urban vegetation. Landsat and Sentinel-2 imagery processed on Google Earth Engine (GEE) is used to derive vegetation indices such as the Normalised Difference Vegetation Index (NDVI), classify green and non-green surfaces, and compute year-wise green-cover area and percentage. The study area is divided into zones or grids to generate green-cover gain, loss and stable maps and to identify hotspots of vegetation decline and recovery. The system further relates green-cover change to built-up expansion and land use, measures green-space fragmentation using landscape metrics, and examines the spatial relationship between vegetation and Land Surface Temperature (LST). All results are delivered through an interactive dashboard with a historical timeline, change-detection layers, hotspot maps, graphs and zone-wise statistics. An optional Machine Learning (ML) module estimates future green-cover trends from the historical series. Overall, the project converts freely available historical environmental data into practical urban green-cover intelligence for planners, municipal bodies and researchers.

Introduction

Urban green cover, comprising trees, parks, gardens, street plantations and other vegetated surfaces, is a key component of urban environmental quality. Rapid urbanisation converts vegetated land into roads, buildings and other impervious surfaces, and the loss is often noticed only after its effects, such as higher local temperatures and waterlogging, become visible. Satellite remote sensing offers a consistent, repeatable record of the land surface over several decades, and cloud platforms such as Google Earth Engine (GEE) now make long time-series analysis practical without heavy local infrastructure [1], [2]. Studies that combine vegetation indices, land-cover classification and surface temperature have shown that vegetation loss and built-up growth are accompanied by rising heat [3].

This project develops the Urban Green-Cover Intelligence and Historical Change Analysis System, a software-based Geographic Information System (GIS) that treats green cover as a time-dependent quantity rather than a static map. The system is independent of any particular city: the study area is supplied as a boundary layer, so the same workflow can be applied to a city, a district or a whole state. Bengaluru and the wider Karnataka region will be used as the initial test area to validate the system.

Motivation

Cities across the world are growing faster than their green infrastructure is being planned, and urban heat, poor air quality and flooding are visible consequences. Planners and citizens usually see only the present green cover and have little evidence of where, when and how quickly it has been lost, or which areas have recovered. Two developments make a systematic historical analysis possible today: the free availability of decades of Landsat and Sentinel-2 imagery, and cloud-based processing that removes the need for expensive hardware [1]. Recent studies have demonstrated time-series green-space mapping and the link between vegetation, built-up growth and temperature [2], [3], but they usually stop at maps or reports. This project is motivated by the need for an integrated, reusable and interactive system that turns such historical data into clear zone-level evidence to support tree planting, green-space protection and sustainable urban planning decisions.

Problem Statement

Existing approaches to urban greenery assessment have the following gaps and limitations:

1. **Snapshot-based mapping:** Most tree and vegetation maps and inventories represent a single date, so gain, loss and recovery over time cannot be measured or visualised.
2. **Lack of local detail:** City-wide averages hide where the loss has occurred, and zone- or grid-level statistics and hotspot identification are rarely provided in a standard form.
3. **Isolated analyses:** Green-cover area is seldom analysed together with built-up growth, land use, fragmentation and surface temperature, even though area alone does not show whether green space is being broken into smaller, less useful patches [4].
4. **Limited usability and foresight:** Multi-year analyses are affected by cloud cover, seasonal vegetation changes and differences between sensors [2], and their outputs are mostly static reports that need GIS expertise to reproduce, with no interactive exploration or trend-based projection for planners.

Statement: To develop a **working web-based GIS platform** that integrates multi-year satellite data to monitor, analyse and visualise urban green-cover changes. The website will provide interactive maps, zone-wise statistics, hotspot detection, fragmentation and LST analysis, along with historical comparisons and optional future green-cover forecasting to support urban planning and decision-making.

Objectives

The project aims to achieve the following specific and measurable objectives:

1. To build an automated pipeline that acquires, cloud-masks and harmonises Landsat and Sentinel-2 imagery for at least five time points spanning two decades or more.
2. To compute year-wise green-cover area (in hectares) and percentage and to generate green-cover gain, loss and stable maps, with a classification overall accuracy of at least 85% and a kappa coefficient of at least 0.80 against reference samples.
3. To divide the study area into zones or grids, rank them by green-cover change, identify statistically significant loss and recovery hotspots, and quantify the relationship of green-cover change with built-up expansion, fragmentation metrics and Land Surface Temperature (LST) using Pearson's correlation coefficient (r) and the coefficient of determination (R^2).
4. To deliver an interactive GIS dashboard with a timeline, change-detection layers, hotspot maps, graphs and zone-wise statistics, and optionally to forecast green cover for the next five to ten years, reporting the Root Mean Square Error (RMSE) on held-out years.

Sustainable Development Goal (SDG)

SDG 11 – Sustainable Cities and Communities. The project is directly connected to Target 11.7, which calls for universal access to safe, inclusive and accessible green and public spaces, and to Target 11.3 on inclusive and sustainable urbanisation and participatory planning. By measuring how urban green cover has changed, where it is being lost and how fragmented it has become, the system gives city authorities and planners the evidence needed to monitor green space, protect it and plan new green areas as cities grow. Its zone-wise outputs also help prioritise the areas where greening is most needed, such as heat-prone zones.

Expected Outcome

At the end of the project, the following are expected:

1. A working, reproducible software system that builds a multi-year green-cover dataset and change maps for any user-defined study area.
2. Year-wise green-cover area and percentage statistics, gain, loss and stable maps, and zone-wise rankings with hotspot maps for the initial study area.
3. Analytical findings on how green-cover change relates to built-up expansion, fragmentation and surface temperature, presented as graphs and coefficients.
4. An interactive dashboard, supported by documentation and a project report, with an optional forecast of future green cover, that planners, students and researchers can use to support evidence-based greening decisions.

References

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